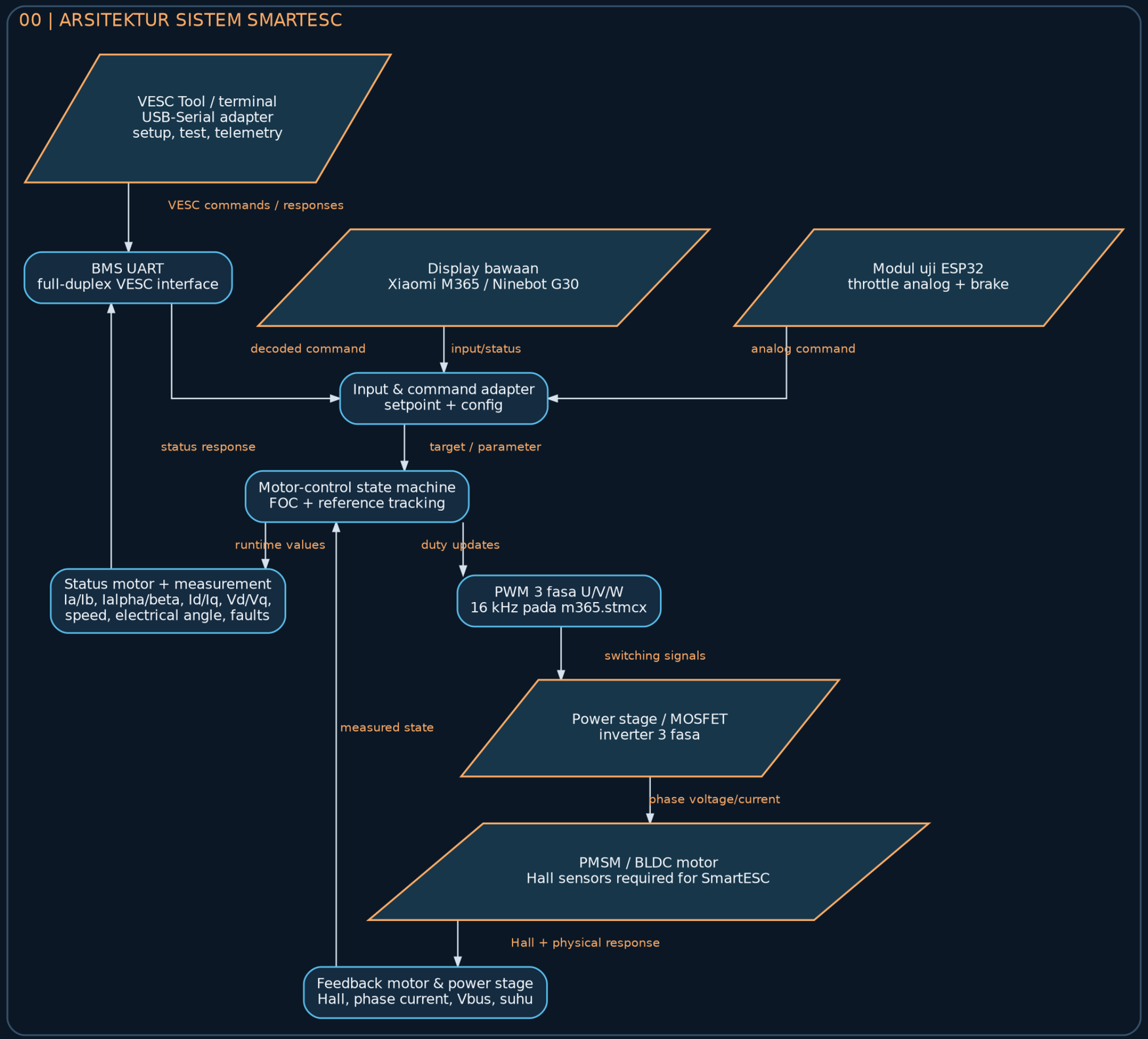
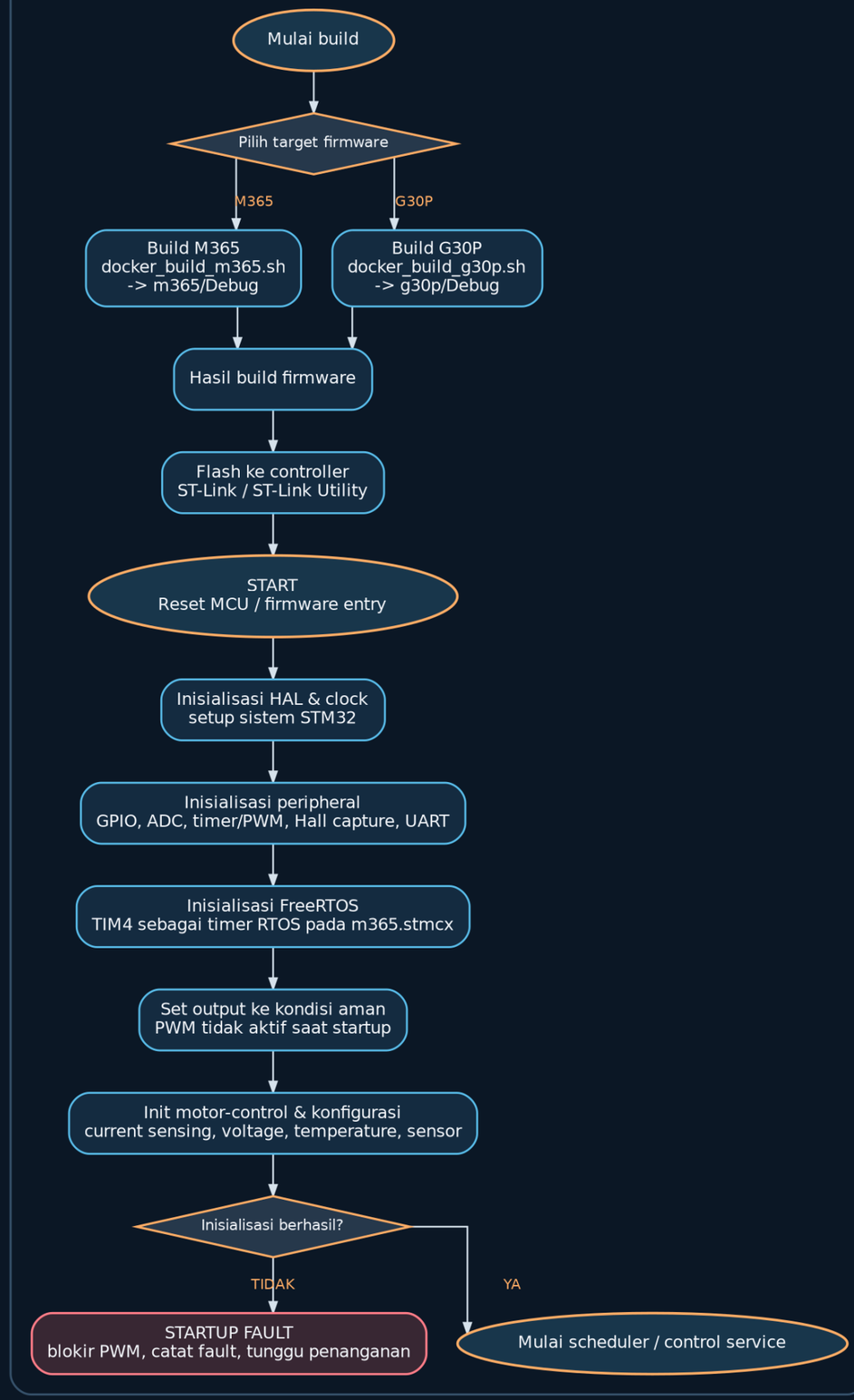
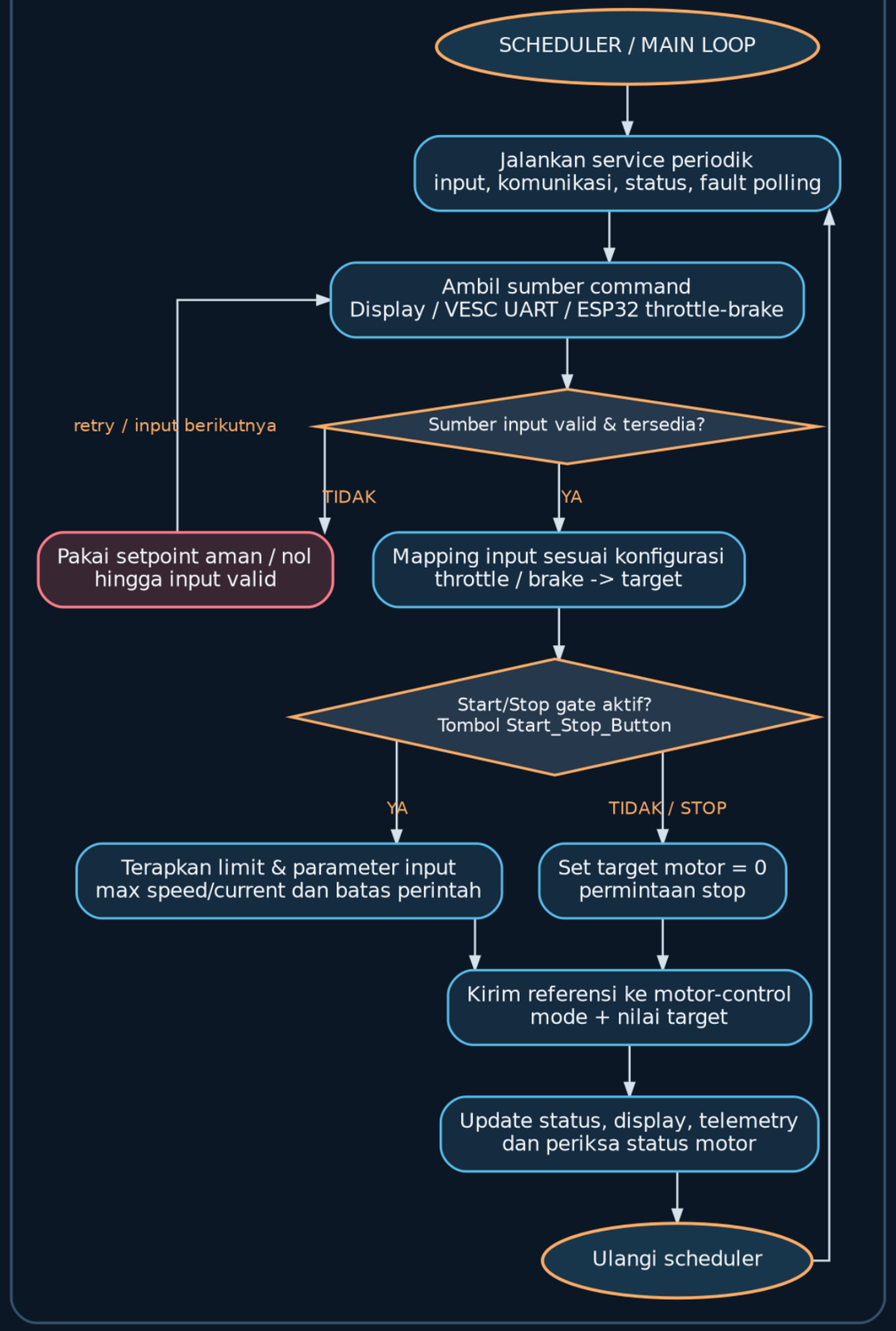

00 | ARSITEKTUR SISTEM SMARTESC
VESC Tool / terminal
USB-Serial adapter
setup, test, telemetry
VESC commands / responses
BMS UART
full-duplex VESC interface
Display bawaan
Xiaomi M365 / Ninebot G30
Modul uji ESP32
throttle analog + brake
decoded command
input/status
analog command
Input & command adapter
setpoint + config
status response
target / parameter
Motor-control state machine
FOC + reference tracking
runtime values
duty updates
Status motor + measurement
Ia/Ib, Ialpha/beta, Id/Iq, Vd/Vq,
speed, electrical angle, faults
PWM 3 fasa U/V/W
16 kHz pada m365.stmcx
switching signals
measured state
Power stage / MOSFET
inverter 3 fasa
phase voltage/current
PMSM / BLDC motor
Hall sensors required for SmartESC
Hall + physical response
Feedback motor & power stage
Hall, phase current, Vbus, suhu

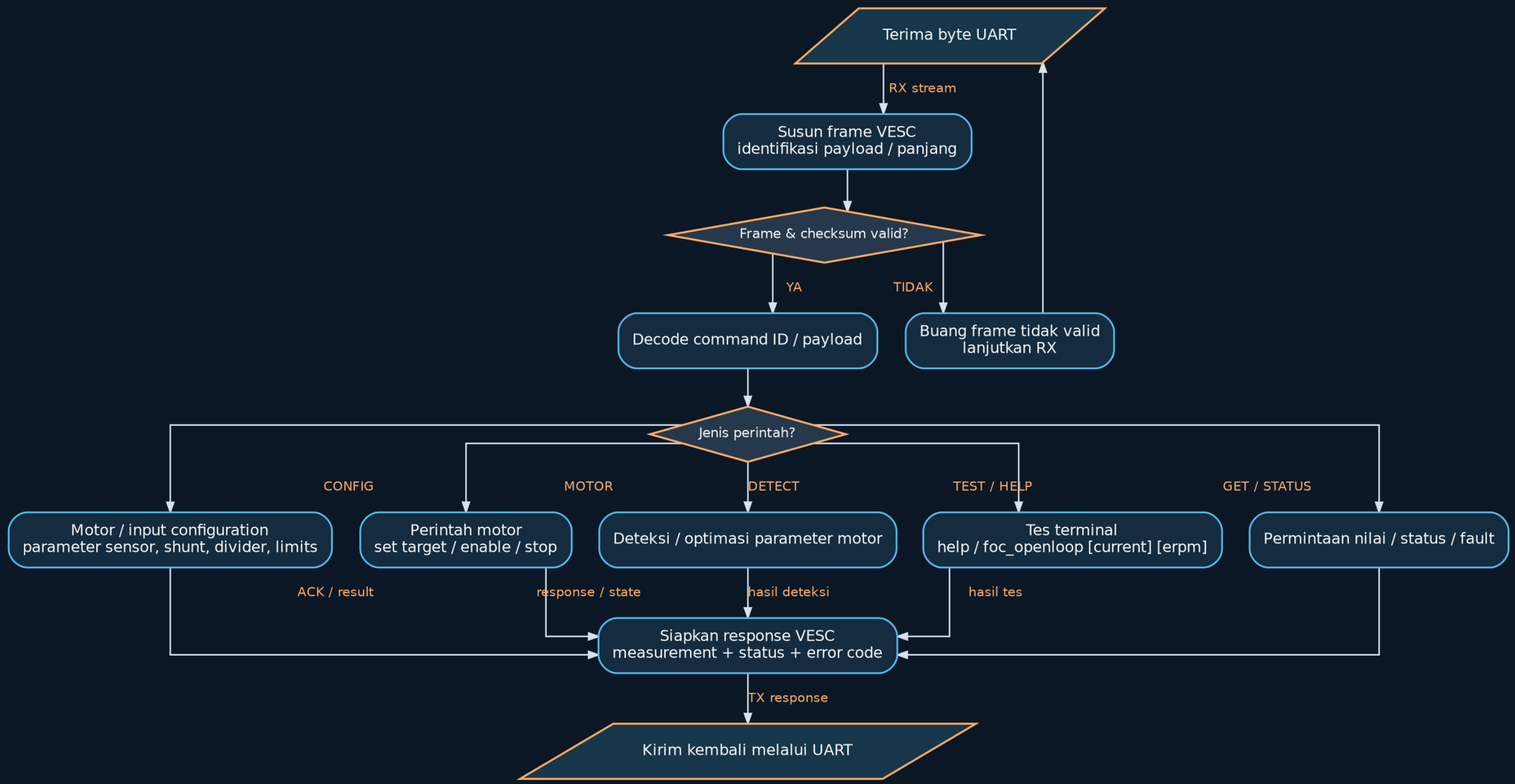
01 | BUILD, FLASH & SYSTEM INITIALIZATION



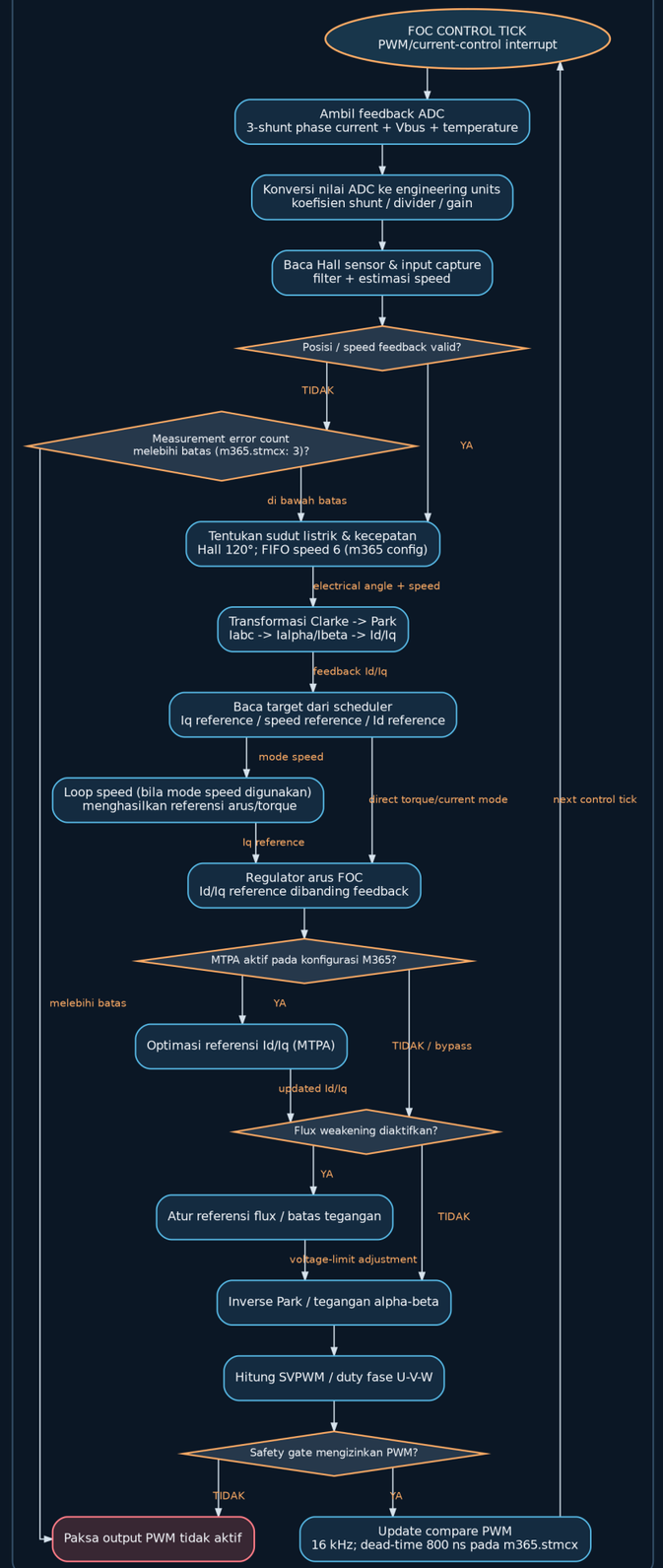
02 | MAIN LOOP, INPUT PROCESSING & START/STOP



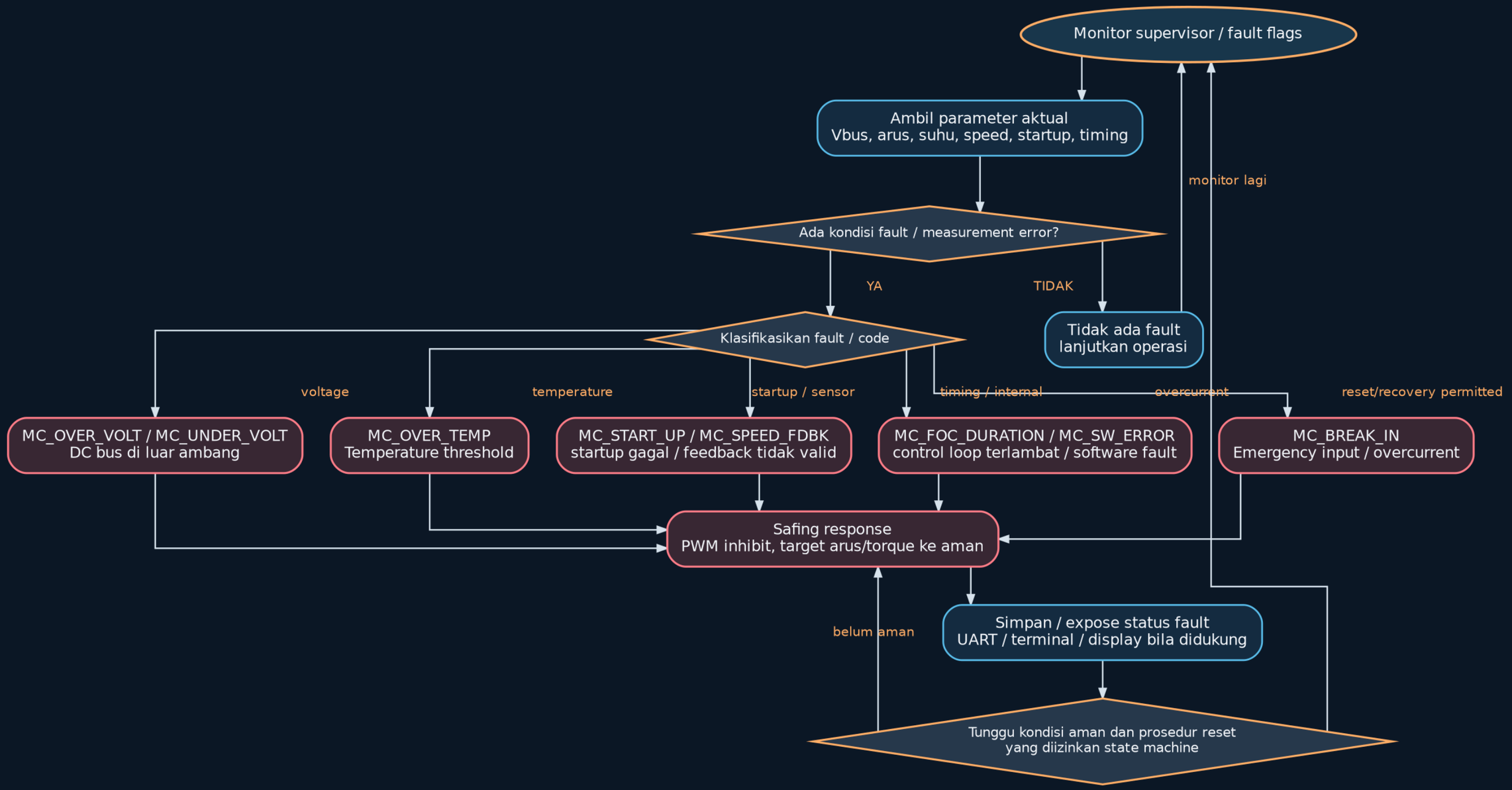
03 | SERIAL VESC, COMMAND DISPATCH & TELEMETRY



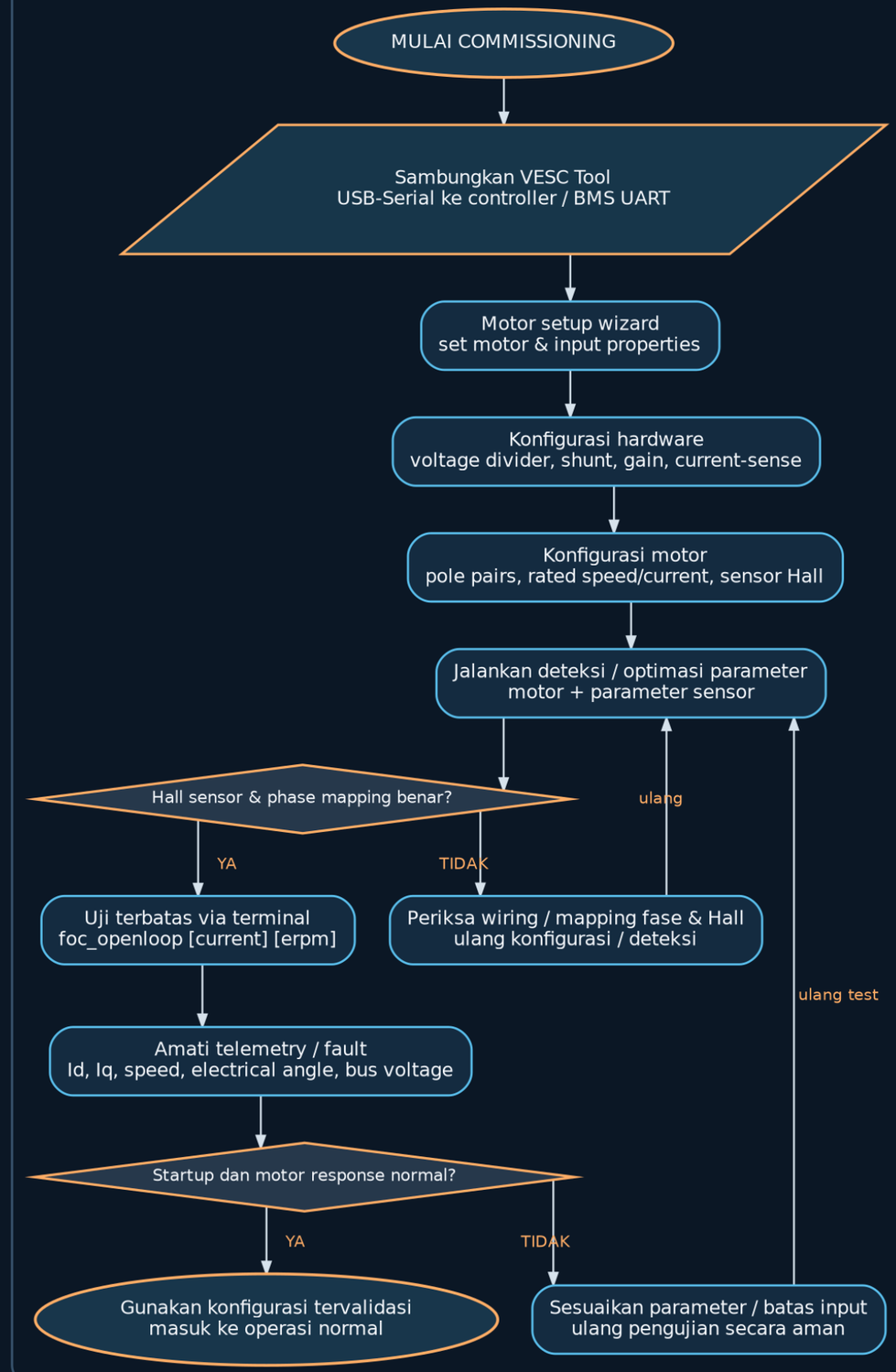
04 | HIGH-FREQUENCY FOC CONTROL LOOP



05 | SAFETY SUPERVISION, FAULTS & REPORTING



06 | MOTOR SETUP, DETEKSI PARAMETER & TESTING



07 | PARAMETER PENTING & BATAS FITUR (m365.stmcx / README)

- README SmartESC: Hall sensors diperlukan
  Sensorless mode belum tersedia
  BMS monitoring dan Bluetooth firmware update
  belum didukung pada status yang didokumentasikan
- MTPA = enabled pada m365.stmcx
  Flux weakening = disabled pada m365.stmcx
  OCP hardware option: disabled pada konfigurasi ini
- Temperature sensing enabled
  70 C threshold; hysteresis 10 C
- PWM 16 kHz; dead-time 800 ns
  Hall speed FIFO depth 6
  measurement errors before fault = 3
- PROFIL M365 (file m365.stmcx)
  15 pole pairs; 2500 rpm max rated speed
  15 A nominal current; 42 V nominal DC
  3 shunt x 0.002 ohm; amplifier gain 8